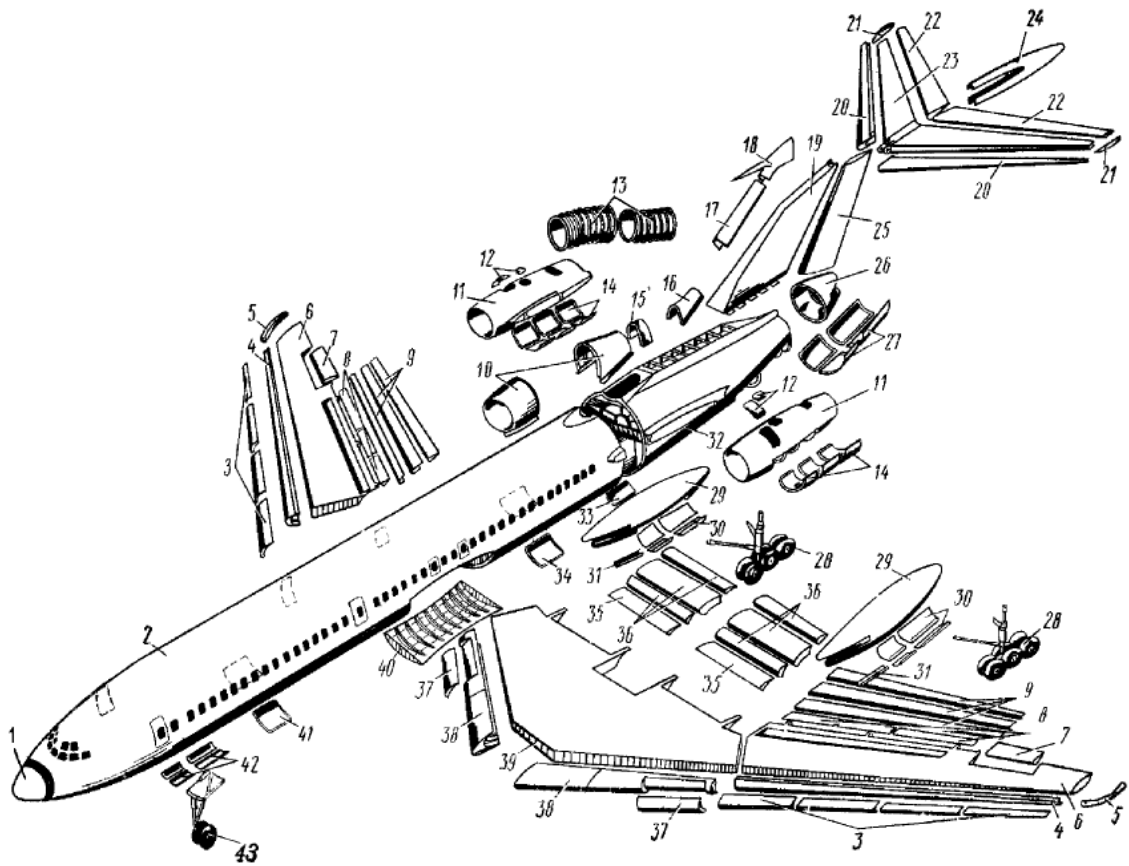


Tu-154B-2 Flight Model and Systems Update for X-Plane11



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Foreword

The creators of the B-2 mod did an outstanding job creating a B-2 version from the original Felis M version. While the visual representation, in particular the cockpit, is arguably as close as currently possible to its real counterpart, some B-2 specific systems and the overall flight dynamics do not match the real aircraft that well, however. This is mainly due to the B-2 mod developers not having access to the original model's source code at the time of the mod's creation and flaws within Felis' base model itself. The shortcomings of the B-2 mod range from small things like some switches, which are not found in the M model, not being properly integrated into the core code to substantial issues with the flight model (e.g. insufficient elevator authority at forward center of gravity). Since Felis made his Tu-154 code, on which the mod is based, openly available, it became possible to remedy most of the mod's issues regarding systems modelling and flight dynamics, which this set of modifications attempts. Moreover, some bug-fixes and additional functionality were added.

Installation

This is NOT a standalone package, you NEED the latest b-2 mod (v 4.1), which can be downloaded from Google drive or Yandex (links can be found on Felis' discord or at <https://vk.com/maestroskaya>).

After installing the original B-2 mod, copy the content of the 'Tu154_b2' folder in the archive into the 'Tu154_b2' (or whatever you have named it) folder in your aircraft folder (e.g. 'C:\X-Plane 11\Aircraft\Laminar Research\Tu154_b2'). Overwrite when prompted.

QuickStart

For those not fancying a read through ten pages on what has been changed/improved, here are the main differences to the original B-2 mod:

- More realistic flight dynamics in pitch. Elevator authority is now sufficient to land and take-off with max forward center of gravity.
- New engine performance model with more realistic fuel consumption at intermediate power settings and more realistic response times.
- This aircraft is much more sensitive in pitch than the original model. I strongly recommend adding curves to your elevator channel.
- Dozens of small bug fixes
- A new VR config file, correcting cockpit seat positions in VR and making most turn knobs (e.g. frequency selectors, temperature selectors, light intensity), turnable with VR controllers instead of the default drag-axis functionality.

Aircraft Systems Changes Overview

Nearly every aircraft system has been revisited/modified to better match real world data/behavior. Some of these are large reworks while others are minor tweaks. Below is an overview of the changes implemented.

Electric System

The power consumption of nearly every single piece of electronic equipment (including the cabin lights and galley equipment) has been changed to match real world data. This means that you can now perform the ABSU preflight checks without overloading the APU-generator/GPU (110 Amps max).

Added left and right AC essential buses. These power half of the fuel transfer pumps, three of the four feed pumps and the left transformer-rectifier. The AC essential buses are normally powered by AC buses no.1 and 2. However, should the AC power distribution system fail while the generators are still working, they can be manually connected to the AC essential buses with the three red guarded switches (one for each gen) to the right of the RA-56 panel. This will restore fuel flow to the engines as well as DC power supply, greatly expanding the crew's options regarding a safe emergency landing.

The B-2 specific transformer rectifier (TR) logic has been merged into the core SASL code. The backup TR will come on automatically in flight should any of the main TRs fail. The backup TR cannot be activated on the ground, except with the black guarded test switch. Depending on the position of the test switch, the backup TR will feed from AC bus no.1 (switch down) or AC bus no.3 (up position). The respective label on the engineer panel has also been corrected (from 1 – 2 to I-III).

Fuel System

Each transfer pump is now individually accounted for. This was necessary to split the pumps fed by the essential AC buses from the rest.

Hydraulic System

Pressure drop with control surface movement has been adjusted according to (scarce) real world data. Nosewheel steering will now consume no.2 hydraulic pressure. Make sure to turn on the no.2 electric pump when taxiing with engine no.2 off (as is common after landing IRL).

Care must be taken when flying with only electric pumps on (e.g. after a triple engine failure). Extending the landing gear while rapidly moving the control surfaces may critically drop hydraulic pressure and substantially degrade controllability.

Air Conditioning and Pressurization

Implemented B-2 specific switches into core SASL code. Unlike the M, the B-2 has separate controls for heat exchangers and heat exchanger-turbines. Since the turbine sits behind the heat exchanger, shutting off flow to the heat exchanger (i.e. full-hot) will render the turbine ineffective. Thus, the heat exchanger is always set to full cold on the ground and the temperature is adjusted using the heat exchanger turbine setting. If the air is not warm enough with the turbine in full-hot (e.g. when it's very cold outside), the heat exchanger control may be used to achieve hotter temperatures. The automatic temperature control will now also adhere to this principle of operation.

The automatic temperature control switch needs to be on and 115V AC power available for the automatic temperature control to work.

The rate at which the cabin heats/cools down has been drastically reduced to achieve more realistic heat up/cool down times. Up to 30 minutes may be required to heat up the cabin from

-20 °C to +20 °C. Open doors will significantly reduce the heating/cooling rate. Activating the outflow valve ejection system will slightly improve heating/cooling rate with open doors.

The amount of airflow from the APU/engines to the bleed air system will now depend on outside air temperature (OAT). Thus, you might want to leave the APU on for taxi on a hot day as the engines alone won't provide enough airflow to keep the cabin cool.

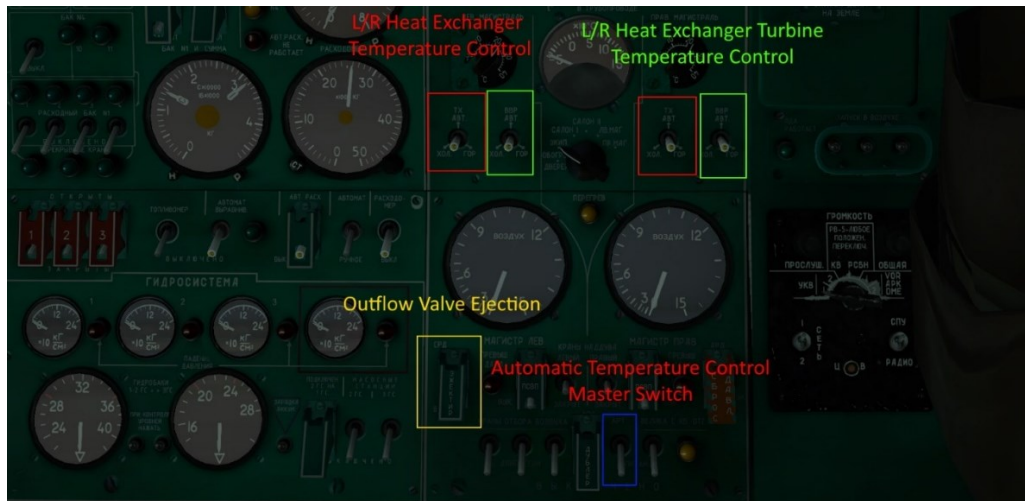


Figure 1 Air Conditioning Panel

Radios

NDB and VOR station morse codes are now audible through the intercom. The original version uses default X-Plane datarefs and sound for the morse codes and it SHOULD work. However, for reasons beyond my understanding it does not for me and a number of other users. Therefore, I implemented the functionality through SASL. Also, reintroduced ADF manual mode. To use it, rotate the mode switch on the ARK panel fully clockwise, set the intercom to ARK-1 or 2 and put the respective 'УЩДБ и СПУ' to 'АРК'. Now press and hold the small red button labeled 'рамка' on the ARK panel. This will rotate the receiver antenna. As the antenna starts to face the station, the static noise will fade and the morse code becomes audible. When the morse code can be heard clearly, the antenna points directly towards the station (or directly away from it!). This way station bearing can be determined manually.

Engine Start System

The conditions for starter activation have been slightly changed. Now, the fuel system does not need to be in 'auto' mode, having the automatic fuel transfer system and the four feed pumps on is enough. For engine no.3, the rear baggage door must be closed. The starter torque value is now plugin controlled and provides for realistic spool-up times. The setting in the EFB is ignored.

Flight control system

The elevator input is now symmetric, meaning that the ratio between physical yoke (or joystick) deflection to virtual yoke deflection will be constant. This was not the case in the original model, where that ratio would change depending on whether you are pulling or pushing your physical yoke. This is a result of controlling an asymmetrical elevator deflection range (29° nose up vs 15° nose down) with a symmetrical physical input device, which ranges from -1 to 1. In practice,

your control sensitivity changed by almost 50% depending on whether you are below or above the neutral point of your yoke/stick. The new model keeps the control sensitivity constant. The downside is that about 25% of your physical travel cannot be used, as the virtual yoke in the Tu-154 will hit its authority stop at around half nose down hardware yoke travel.

Fire Protection System

The fire protection system has been integrated into SASL code and is now fully functional within X-Plane's default fire suppression logic. Once fire agent has been introduced into an engine, it cannot be restarted, however. This is not 100% true to real life where an engine can be restarted after windmilling/cranked fire agent out of the engine, provided the engine is undamaged. This may change in the future.

APU

APU spool-up forces have been tweaked for a more realistic acceleration during startup. The new model will also allow APU starts at very low temperature (down to -40°C). Previously, the APU would not start below -25°C. APU oil and exhaust gas temperature models also saw a complete rework for more realistic values depending on OAT and APU load.

APU rpm will now also react to electric and pneumatic loads. When bleed air is drawn from the APU and/or the APU generator is turned on, rpm will drop momentarily by a few percent before getting restored to the nominal value, which itself is a function of APU load and may be as high as 101% (as opposed to 99% in idle). This is most notable during engine start, where the equivalent of approx. 300hp is drawn from the APU.

Finally, APU compartment heat has been implemented. Hot air from engine no.2's compressor is routed into the APU compartment whenever compartment temperature drops below 0 °C. This facilitates APU in-flight start by keeping the APU's regulator pump, combustion chamber and oil at reasonable temperatures during cruise flight, where OAT can be as low as -62 °C. The system is inhibited on the ground and whenever the APU is running. APU oil temperatures of approx. 50 °C in cruise indicate a working APU compartment heating.

EGT Protection System (RT-12)

Also integrated into SASL and fully functional beyond the test procedure. The RT-12 is active during startup, reverse thrust and above 4000 m altitude. It will reduce engine fuel if EGT rises above 650 °C and cut the fuel above 710°C. During startup, these thresholds are reduced to 540 °C and 600 °C respectively. The RT-12 disengagement switches (red guarded left of the flight recorder panel) are now present on all three versions. Previously, they were omitted on the 90s panel. However, most aircraft had them refitted, so I added them into the 90s-panel version.

Automatic Flight Control System (ABSU)

The ABSU saw a large rework. The manual flight control logic has been completely reworked, implementing real-world control laws. For roll and yaw, this includes bank/yaw acceleration damping with a low-pass filter. For pitch, pitch acceleration damping and the longitudinal control mode (LCM)¹.

¹ This is not an official designation but rather something I have come up with. The original Russian term is 'Продольная управляемость', which translates to 'longitudinal controllability'.

What is 'longitudinal control mode'? Even back in the day, there were certain requirements on how large airliners had to handle in pitch and bank. For pitch, there was a particular requirement for a minimum amount of yoke force/yoke deflection needed to generate one G of vertical acceleration. This so-called control characteristic required no less than 50mm of yoke travel and no less than 10kg of force to generate one G. Moreover, the control characteristic had to be constant through the flight envelope. With a bit of math and aerodynamical theory, one will find that the amount of elevator deflection needed for one G of vertical acceleration will vary substantially with aircraft center of gravity (CG) and airspeed (the more forward the CG, the more elevator will be needed). For an aircraft like the Tu-154, which has a large range of permissible CG, this posed a problem since the required control characteristic could not be achieved with mechanical controls only. To solve this problem, the LCM was implemented in the ABSU. The LCM increases or reduces pilot input to the elevator based on aircraft CG. If the CG is far forward, elevator deflection is increased by adding nose up input through the autopilot's pitch servos. If the CG is far behind, elevator deflection is reduced through the ABSU servos. Since the Tu-154 does not have a CG computer (probably because it was deemed to expensive), a simple yet elegant solution was found: The elevator trim position was used to determine how much elevator deflection the ABSU should add to pilot input. This works well provided the aircraft is trimmed hands-off in level flight since a nose-heavy aircraft will require a lot of nose-up trim while a tail-heavy aircraft might even require nose down-trim. Moreover, this method accounts for airspeed changes. With this additional input through the ABSU's LCM, the required characteristic could be achieved within a reasonable margin, both in X-Plane and the real aircraft.

The above also highlights why it is imperative to have the aircraft trimmed for hands-off level, flight at all times, since the trimmed position directly influences yoke-to-elevator ratio and thus its longitudinal flight characteristic.

Another small yet important change was made to the roll-mode. Whenever the TKS realignment switch is pressed (either to the left or right), the current heading becomes the desired heading that the ABSU will try to hold in STAB mode. This means that the aircraft will no longer turn in STAB mode when re-aligning the gyros prior to top-of-descend. Another small yet convenient change is the ability to bring up nav indications (either NVU or VOR1/2) on the captain's HSI without having to turn on the NAV PREPARE switch. Thus, you can fly the aircraft in roll/pitch stab while changing HSI indication.

A final note regarding the yaw channel. The ABSU's yaw channel is rather primitive. It will deflect the rudder opposite to current yaw-acceleration. This means that it will not facilitate coordinated turns. This does not mean however that Tu-154 pilots put in bottom rudder for turning. Instead, they would simply accept a small deviation of the ball during the initial phase of the turn.

When using the autoland system, a special yaw-rate damping mode is activated once the glideslope (GS) mode is activated. This means that you have to be dead-on runway-centerline before activating GS mode. Otherwise, the roll channel will fight the yaw channel, the former trying to turn the aircraft on the centerline and the latter trying to stop any turn.

Autopilot Control Servos (RA-56)

The autopilot control servos are now modelled in more depth, replicating some of the intricacies of the ABSU.

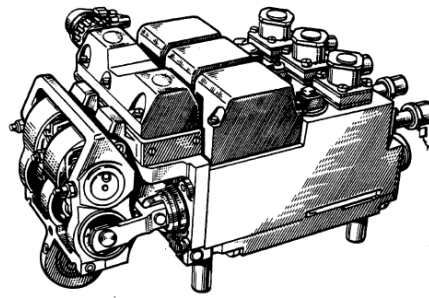


Figure 2 RA-56 Control Servo

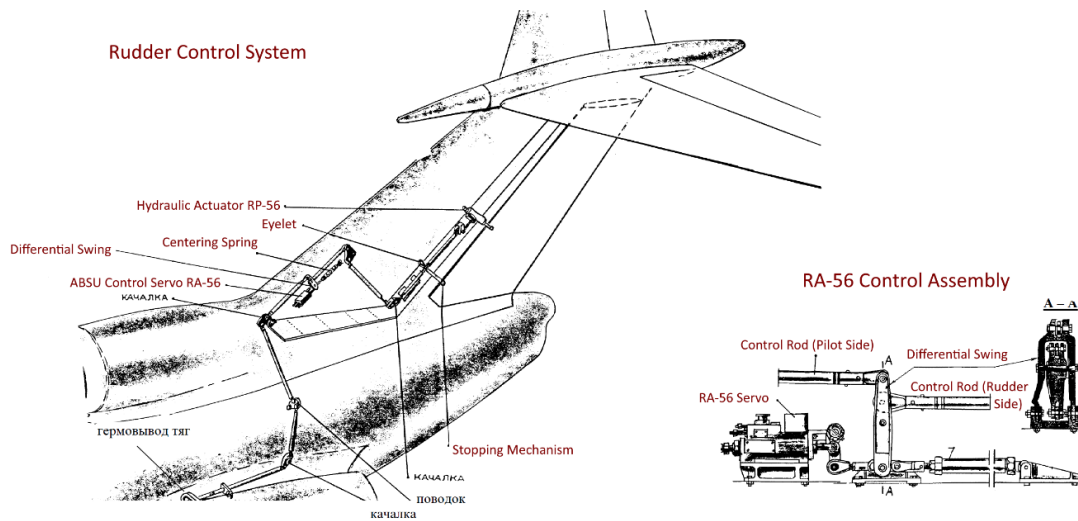


Figure 3 Rudder Control System and RA-56 installation. (Source: В.И.Зинченко; КОНСТРУКЦИЯ И ЭКСПЛУАТАЦИЯ САМОЛЕТА ТУ-154Б)

The ABSU controls the aircraft's flight control surfaces through a set of hydraulically powered control servos, designated RA-56. They are connected to the mechanical control rods through a special mechanism that adds pilot and servo control inputs (so-called differential swing).

Each servo is powered by three independent hydraulic actuators, fed by no.1, no.2 and no.3 hydraulic systems respectively. A total of nine such RA-56 hydraulic actuators are installed on the aircraft (three for each channel: pitch, yaw and bank). Hydraulic power supply to these is controlled from the RA-56 panel (see figure four). A yellow lamp below each switch indicates removal of hydraulic power from the respective actuator either manually through the switch or by the automatic shutoff system.

The automatic shutoff system monitors actuator position and removes hydraulic power should one actuator's position differ from the other two. This can happen either due to a failure in the control feedback signal (which drives the affected actuator to the fully extended position) or from insufficient hydraulic power in the respective hydraulic system. Should all three actuators show different positions, hydraulic power is removed from all actuators, rendering the entire channel inoperative. This can also happen during initial activation of the ABSU. While the aircraft is powered down, hydraulic pressure bleeds slowly from the hydraulic systems and actuators. This happens at different rates and may result in the actuators moving to different positions over time. Now, when the ABSU is powered on after the aircraft was left unpowered for

a longer time, the automatic shutoff system will come online and may detect actuator position discrepancy faster than then actuators can return to the neutral position. This will trigger a shutdown of the entire channel as described above. Thus, the automatic shutoff system needs to be switched off when the ABSU is powered on for the first flight of the day. To do so, put the automatic shutoff switch (marked in figure four) in the up position before turning on the ABSU. Put it back down once the green ABSU ready light comes on on the PPN panel (the one with the Greek letters to the right of the RA-56 panel).



Figure 4 RA-56 Panel

Engines (NK-8-2U)

Engine performance saw a complete rework in the form of an entirely new model built from scratch. The original model, while reasonably good within what was possible without access to the source code, was rather inaccurate in some areas. In particular, engine response was too fast and fuel consumption at intermediate power settings was too low (by about 50%)

The new engine model features accurate fuel consumption, N1/N2, EGT and thrust in a wide envelope, from -40 to +40, sea-level to operational ceiling (12100m or FL40). Moreover, engine spoolup/spooldown as well as startup and rotor runout times more closely match real world data. The effect of the high-pressure compressor bypass valves is now also modelled. The bypass valves close above 74.5% N2 which rapidly increases compressor efficiency and thrust.

Since the original engine model was close enough for cruise settings, the fuel planner built into the EFB can be used with the new model.

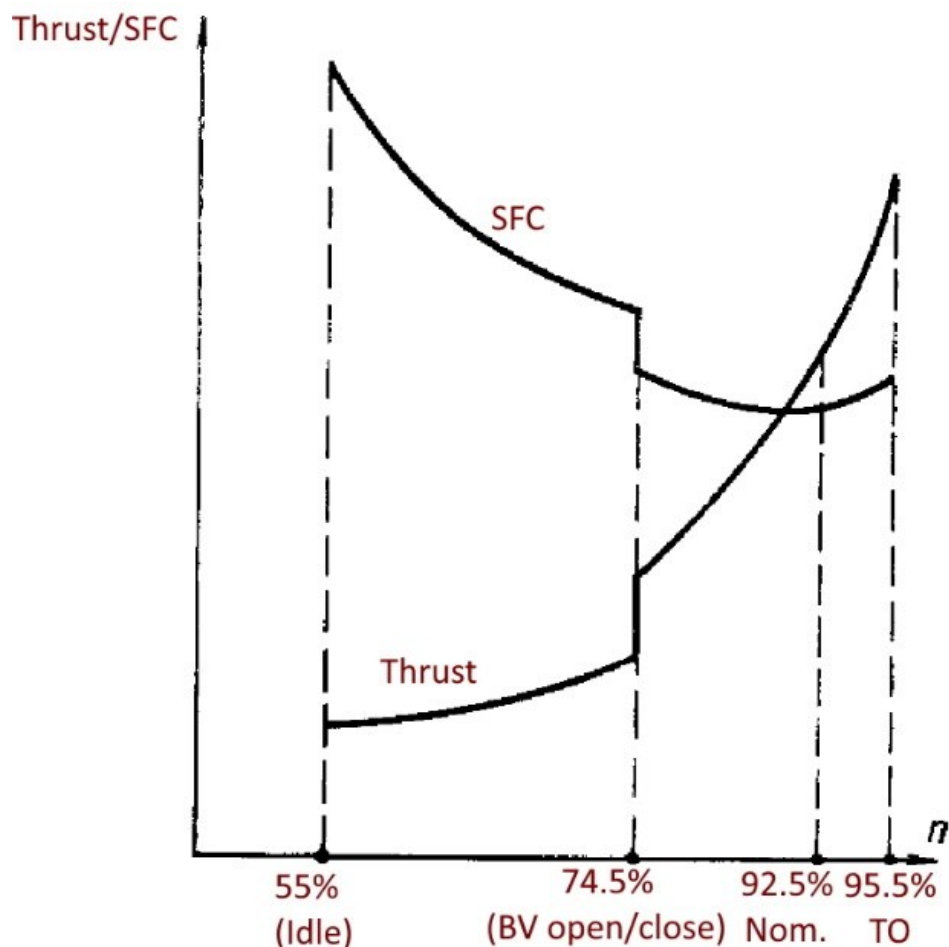


Figure 5 NK-8-2U thrust and specific fuel consumption (SFC) curves (sea level, ISA). BV: Compressor bypass valves, Nom.: Nominal power, TO: Take-off power. Source: Самарский государственный аэрокосмический университет имени академика С.П. Королева: Конструкция и летная эксплуатация двигателя НК-8-2У

Sounds

Added 36V transformer sounds to the cockpit and the characteristic high-pitched internal engine sound of the NK-8 engines. The latter is barely noticeable in the cockpit but very prominent in the back of the cabin. A lot of small sound tweaks and bug fixes have also been made (e.g. wrong audio channel calculation for external engine sounds).

Flight Dynamics Changes Overview

Pitch Authority

Perhaps the biggest issue with Felis' original model and consequently the B-2 Mod is insufficient elevator efficiency. I supposed Felis intentionally tuned it down to make flying the plane more enjoyable with common desktop flight controls. However, this resulted in too much elevator needed for approaches at forward CG settings (sometimes even past the trimmable range) and the inability to lift the nose gear at computed Vr speed for CGs below approximately 25%MAC (the real aircraft's limits are 21% on take-off and 20.5% for landing). A (tedious) rework of elevator/stabilizer effectiveness and flap moments brought the flight-model (FM) much closer to available data on pitch effectiveness. Figure six shows balancing curves (i.e. how much elevator deflection is required to maintain level flight at a specific airspeed and configuration)

from real flight-testing data and a point-wise comparison between the new (green markers) and original (red markers) FM.

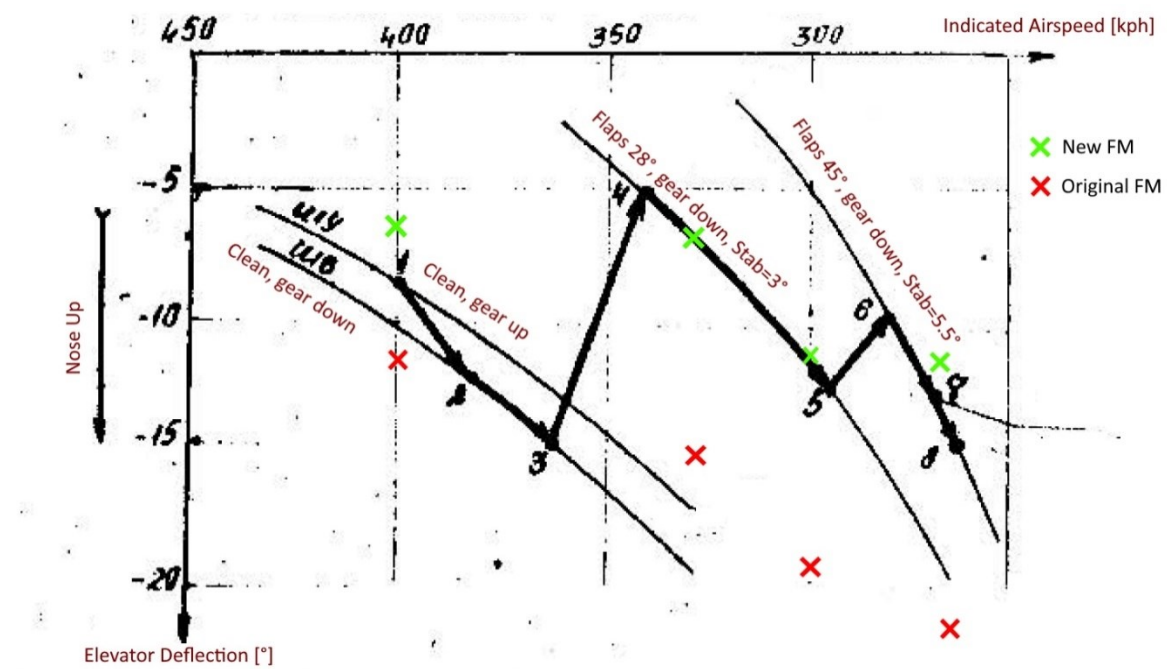


Figure 6 Elevator balancing curves with a point-wise comparison between the new and the original flight model (FM). Aircraft weight: 75t, CG: 18.5% MAC. Source: Г.С. Пуминова: Практическая аэродинамика самолета Ту-154Б(М).

While both models are reasonably close to real world data in a clean configuration at 400kph, the original model diverges heavily from the balancing curves at lower speeds and extended flaps, requiring well above 20° nose up elevator in landing configuration at approach speed (260kph). The new model on the other hand remains close to the balancing curves. Moreover, the aircraft will now rotate at computed V_r speed up to the maximum permissible forward CG of 21% MAC (and even beyond). While not

The new FM, while being substantially more accurate, leads to a strong increase in pitch sensitivity compared to the original FM. On a common desktop yoke, which features around 5-10cm of pitch travel compared to the real aircraft's 40cm, this more accurate FM may be more difficult to fly with compared to the original FM. I therefore strongly recommend using a curve on the pitch axis with an initial ratio of 0.5-0.3.

Ground Effect

X-Plane's FM system does model the ground-effect induced change in lift, drag and pitching-moment. It grossly exaggerates these changes however, resulting in too much lift and not enough drag close to the ground. This makes the aircraft more prone to 'floating' during the flare maneuver, much more than in real life. Felis did try to remedy this by reducing lift as a function of altitude above ground which was a substantial improvement over the default behavior provided by X-Plane but was still off lift and drag-wise. The new FM tunes lift in drag close to the ground even further and eliminates the pitch-down moment, resulting in more realistic flare dynamics. In particular, the new FM can replicate the following behavior observed on the real aircraft:

“To evaluate the effect of ground proximity on the aerodynamical characteristics, a special approach was performed with no control inputs from the pilots with an initial vertical speed of 2 m/s. With constant thrust and no control inputs from the pilots, the aircraft, trimmed for descend, will stabilize approx. 1m above the ground while retaining its speed”²

All in all, the new FM makes soft and controlled landings easier and behaves more in-line with what is described by real pilots.

Thrust Dependent Pitch-Moment

The real aircraft exhibits a noticeable thrust-induced pitching moment in landing configuration. Reducing thrust from approach power to idle corresponds to shifting the CG by approximately 2-3% forward. This means that reducing power will require more nose-up elevator to maintain your flight path. Therefore, reducing engine rpm below 75% on the glide is not advised, especially at forward CG settings. In a worst-case scenario, e.g. when pulling out of a steep descend with engines at low rpm, elevator authority may be insufficient to generate enough pitch rate to prevent ground collision (see Aeroflot 3603 for tragic example).

Weight and Balance

Aircraft empty weight and longitudinal positions of fuel tanks, seat rows and baggage compartments have been adjusted to real world data for the 164-seat configuration. A slight change of aircraft CG with gear extension/retraction is modelled. Empty operating weight is now 53t instead of 51t, assuming four crew and four flight attendants. The ‘equipment’ weight setting in the load manager can now only be set to 0 or 370kg, which corresponds to center-galley equipment not installed or installed.

Compatibility

The modifications provided here improvements have been tested with the following plugins:

- Xenviro
- BetterPushBack
- Simple Ground Services
- JARDesign Ground Handling Deluxe
- ASXP
- WorldTraffic

I did not test with other plugins so I cannot guarantee that they will not break the aircraft. If you find some interference of any plugin with the aircraft, let me know either through Felis’ discord, which I regularly check, or an X-plane.org forum DM.

SmartCopilot

This version has been made compatible with Smarcopilot 3.3.0. Some things have to be considered, however, when using SC:

GPS Sync: Since X-Plane’s default GPS does not expose its state via datarefs, GPS is not actually synced between master and slave. Only the inputs to the GPS are synced, which means that both players need to have the exact same navigation and flight plan database. Moreover,

² Лигум Т, Скрипниченко С, Шишмарев А: Аэродинамика самолета Ту-154Б, Page 140.

both pilots need to set the same page and cursor state before connecting, otherwise the GPS may become out of sync. In any case, the plane will follow the master's GPS in nav (NVU) mode.

KLN Sync: Unlike the GPS, the KLN is fully synced between master and slave. Both pilots will see what the master's KLN displays. This means that both players do not need to have the same database.

Before connecting, the TABLET MUST BE OFF, otherwise the planes will not sync properly. Once connected, any player can turn on the tablet and set fuel and payload. Avitab features are not synced, however.

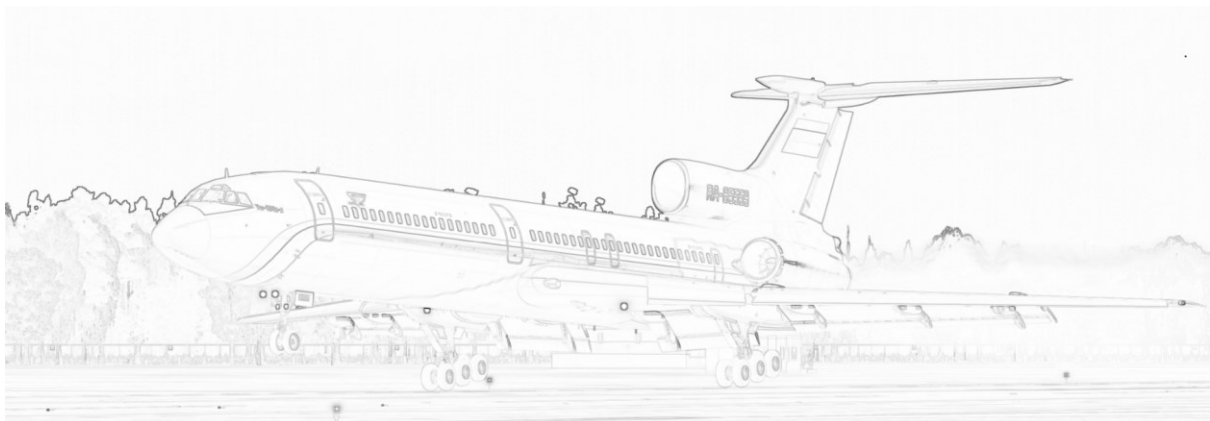
I have found during testing that sometimes SC would not sync properly on the first connection attempt. After connecting, the master should move the yoke back and forth a bit and confirm that the slave also sees yoke movement. If the yoke does not move on the slave's plane, disconnect and reconnect. It may take several attempts to achieve a proper synchronization.

Closing Remarks

This FM and systems update is the result of countless hours of research, coding and flight testing. While I tried to stick as much to real world data as possible, some information was simply not available and educated guesses were made. If you find anything that does not behave like it should, please let me know either through Felis' discord (Tu-154 channel), which I regularly visit, or an X-plane.org forum DM. However, please refrain from posting complaints based on how you feel the plane should fly/behave. If, on the other hand, you have evidence in the form of real word flight-test or systems description data, I'm happy to look into it.

Will the Tu-154M get the same treatment as the B-2? Probably not. Looking back at how much time I have invested in development, I'd rather do more flying instead of coding, especially with a solid Tu-154B-2 now on my hands. I do personally favor the M over the B-2 though, especially with the UNS mod. Therefore, I might find the motivation to update the M someday.

The modifications and improvements provided here bring the already impressive B-2 even closer to the real thing, perhaps as close as is possible in X-Plane. I hope you will enjoy it as much as I do.



Changelog

V 1.2

ADDED:

APU compartment heat (see readme for further details)

IMPROVED:

Aileron effectiveness adjusted throughout the flight envelope to match real-world data

Rudder effectiveness adjusted for proper outboard engine-out behavior (approx. 7-8° of rudder are needed to counter asymmetric thrust on glideslope with flaps 28) and yaw induced roll-rates.

More realistic, custom engine oil temperature model

More realistic APU oil temperature and EGT model

Load-dependent APU rpm (see readme for further details)

FIXED:

Aileron trim not working

Flight controls and NWS working without hydraulic pressure or servo power

APU fire test-logic (circuits 4-6 are not present on the APU)

Bleed duct overheat light not working

V 1.1

ADDED:

Russian Manual

Smartcopilot compatibility

IMPROVED:

Nosewheel steering (NWS) logic changed to be more accurate. For users having a dedicated tiller axis, both rudder pedals and tiller can control the nosewheel in 63° mode (pedals only up to 10°). When in 10° mode, the tiller is locked in the neutral position and NWS is controlled using rudder pedals only. For users without a dedicated tiller axis the logic is the same as the previous one.

Nosewheel turn rate will now depend on nosewheel load, taxi speed and no.2 hydraulic pressure. When using Betterpushback, the NWS system needs to be turned off with the switch on the captain's yoke, otherwise the tug will be unable to turn the nosewheel (the Tu-154 has no bypass pin).

FIXED:

Excessive fuel consumption at high cruise speeds (>0.82M)

Feed tank value becoming NaN (not a number) and thus killing the engines in cruise.

Rapid pitch down below 40m under certain conditions.

APU ready lamp logic. Will now extinguish after the bleed valve is fully open.

STU lamp not illuminating with NAV PREPARE switch on.

Baggage fire pushbuttons not illuminating when pushed.

V 1.01

FIXED:

No.2 engine going to full power when reverse thrust is applied.

V 1.0

Initial release